

Project Plan for an AI Assistant for Vacations for YOU

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This form is completed in accordance with the Software Engineering Body of Knowledge (SWEBOK).

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Introduction

Vacations for You manage a high-volume vacation rental operation and processes a large number of guests inquires through a traditional call center model. This project focuses on the development of a functional Proof-of-Concept system capable of handling common guest inquiries such as check-in/check-out times, reservation details, cancelation policies, property amenities and other routine informational requests. The product will be a web based, AI assisted support chat system to help all guests and potential guests. The system is indented to reduce call-center load by answering common questions and providing a clear escalation path when needed.

Project Overview

2.1 Scope

The document specifies the project plan for the Phase 1 proof of concept AI-assisted guest support system for Vacations for YOU. The system outlined is designed to support guests with existing reservations by providing a fully conversational chatbot experience that delivers automated responses to common post-booking questions through a web-based chat interface.

The Phase 1 system will employ a large language model (LLM) to generate responses based on read only reservation, property, and policy data from the StreamlineVRS property management platform or an approved sandbox environment. The AI assistant will provide informative responses and present clear escalation guidance when questions fall outside of the scope or confidence threshold.

The project excludes booking creation or sales, payment processing, service order placement including but not limited to maintenance, account creation, or authentication beyond what is necessary to provide reservation information. The project is intentionally structured to validate feasibility, usability, and call-deflection potential for future channel expansion.

2.1.1 Identification

System Name: Vacations for You AI Support Assistant

Version: 1.0

Document Type: Project Plan
Sponsor: Vacations for You

2.1.2 System Overview

The AI-assisted guest support chat platform will follow a modular layered, client-server architecture. This separation of components improves maintainability and scalability while remaining appropriate for the POC implementation.

The architecture of the platform consists of:

- lightweight web-based frontend using React
- Python-based backend service using FastAPI
- AI processing layer utilizing a Large Language Model (LLM)
- Integration layer responsible for secure communication with the StreamlineVRS API.

The front-end provides the conversational UI through which guests interact, submit questions and responses. Communication to the back end will be done through HTTPS REST Endpoints. All integration with the StreamlineVRS API is strictly read-only to prevent reservation data from being modified by the system. The backend service will manage session handling and reservation validation. The AI processing component will implement a Retrieval-Augmented Generation (RAG) pattern, through which relevant reservation and property data are retrieved from StreamlineVRS prior to constructing the prompts. These prompts will then be submitted to an external LLM provider, such as OpenAI, to generate NL responses relevant to the guest's inquiries.

2.1.3 Document Overview

This document serves to outline a structured framework for managing and executing the project from initiation through deployment. It defines the project's scope, objectives, timelines, resources, and risks. It serves as a baseline reference for stakeholders and project team members throughout the lifecycle of the system.

2.2 Current System or Situation

Currently, Vacations for You relies on a traditional call center model to handle guest inquiries. This approach requires significant human staffing resources and results in high operation overhead. Many of these inquiries involve repetitive informational questions that could potentially be automated.

2.3 Background, Objectives, and Scope

With the rise of AI, using automated chat bot agents has proven itself for simple repetitive tasks. Several companies are already utilizing AI assistants to quickly provide customers with information and take a hefty workload off of human agents. This project is meant to provide a simpler and more efficient form in which customers can obtain information from Vacations for You.

Objectives:

- Develop POC AI chatbot
- Demonstrate ability to retrieve and present accurate reservation data
 - Validate chatbot efficiency
- Reduce call-center workload
- Offer 24/7 self-service support

2.4 Operational Policies and Constraints

The system must operate under the following constraints:

- No modification of customer data
- No handling of monetary transactions
- Unsupported questions must be escalated to human agent
- All API access must be read-only
- All communication must be secure
- Sensitive customer data must not be stored.

2.5 Users or Involved Personnel

- Guest with existing reservations
- Potential guest seeking property/area information
- Call center representatives
- Project sponsor
- Project development team

Development Background and Approach

3.1 High Level Estimates

The project will be completed within one academic semester and be divided into phases seen in the table below:

Phase	Duration
Requirements & Design	2 weeks

Prototype Development	5 weeks
Testing & QA	2 weeks
Refinement	1 week
Finalization	1 week

3.2 Key Contacts and Stakeholders

- Team Lead: Responsible for all external communication with stakeholders and keeping the development team on track.
- Team Members: Responsible for frontend, backend, and testing assigned tasks.
- Project Sponsor: Provides initial requirements, validation, and feedback on product development.

Features, Primary Deliverables, and External Commitments

4.1 Feature List

The phase one POC AI support assistant will include the following core features:

1. Web-Based Chat Interface
 - Embedded chat widget accessible from any page on existing website
2. Reservation Validation
 - Reservation lookup using reservation ID and/or Last Name
 - Read-long validation against StreamlineVRS or approved sandbox
3. AI-Powered Conversational Responses
 - Natural language understanding of guest inquires and context aware responses based and validated reservation data
 - Multi-turn conversational support
4. Retrieval Augmented Generation (RAG)
 - Dynamic retrieval of relevant reservation, property, and policy data
 - Prompt construction using retrieved data
5. Escalation & Human Handoff
 - Confidence based escalation mechanism
 - Clear instructions for contacting human support
 - Logging of escalation events
6. Conversation Logging
 - Session logging of guest questions and system responses
 - Tracking of unsupported inquires and escalation

4.2 Customer Deliverables

The final deliverables will include:

- Working web-based POC AI chatbot
- Technical documentation
 - Project Plan, SRS, SDD, User Manual
- Evaluation report and metrics
- Presentation and demonstration
- GitHub repository

Project Schedule

5.1 Major Project Milestones

This project is broken into three major milestones that we have defined as follows:

- Milestone One: Documentation Phase ---- 2/18/26
 - During this phase we are focusing on developing Requirements Documentation, and Design Documentation. Our focus will be requirements elicitation and developing a design based on the requirements gathered. Once both documents are approved, we will move forward into basic API implementation using OpenAI to set up milestone two.
- Milestone Two: Prototype Development ----- 3/25/26
 - After completing milestone two, our goal will be to create a working chatbot that can answer the basic questions needed to work as proof of concept. As an initial prototype, we expect to focus on the most basic implementation, and plan on expanding based on the success of the initial chatbot implementation. Documentation will be updated throughout this process.
- Milestone Three: Testing and Refinements ----- 4/22/26
 - During this phase, the Chatbot will be basically functional, and the documentation will be updated to reflect it. At this phase, we will be testing and expanding the chatbot as well as preparing for C-Day and the final presentation.

Project Work and Product Estimates

6.1 Estimate Summary

Project Resource Requirements

7.1 Staffing and Skill Requirements

Since this project is being developed by a student team, staffing will primarily consist of team members taking on multiple roles. The most important technical skills required include experience with API integration, basic backend development, and an understanding of Large Language Model (LLM) implementation. Team members should be comfortable working with REST APIs, handling authentication keys securely, and sending/receiving structured data (such as JSON). Knowledge of prompt engineering and how to structure inputs and outputs for an LLM will also be essential.

In addition to API and LLM integration, the team will need foundational skills in frontend development, version control (e.g., GitHub), and basic database management. Since this is a capstone-level academic project, advanced AI model training is not required; instead, the focus will be properly integrating and configuring an existing LLM service through its API.

7.2 Plan to Fill Skill Gaps

Any gaps in API or LLM knowledge will be addressed through self-learning, online documentation, and tutorials provided by the selected platform (e.g., OpenAI or similar services). The team will review official API documentation, experiment in sandbox environments, and build small test implementations before integrating features into the main system.

If technical challenges arise, the team will seek guidance from the course instructor, faculty advisors, or reputable developer communities. By dividing responsibilities and collaborating closely, team members can support each other in areas where one person has stronger experience, ensuring successful implementation of the API and LLM components.

Dependencies and Constraints

8.1 Constraints

The development and evaluation of this system are subject to the following constraints. The project must be completed within the duration of a single academic semester. No live customer or order data may be accessed, stored, or processed at any time. System integration with sponsor platforms shall be limited to read-only access, whether through live endpoints or approved sandbox environments.

All changes to the defined project scope must be formally approved by the sponsor and documented prior to implementation. These constraints are intended to reduce risk, protect sensitive data, and ensure that the project remains aligned with sponsor expectations and academic requirements.

Risk Management

9.1 Risk Management Strategy

Risk management for the Vacations for You AI Support Assistant will follow a proactive and continuous monitoring approach throughout the project lifecycle. Because this is a semester-based Proof of Concept (POC) project involving API integration and Large Language Model (LLM) services, risks will be identified early, reviewed regularly during sprint meetings, and documented in a shared project tracking document.

The team will evaluate risks based on likelihood (Low, Medium, High) and impact (Low, Medium, High). Risks rated as Medium or High impact will require mitigation strategies and a designated team member responsible for monitoring them. Risk reviews will occur at the start of each sprint and prior to each major milestone to ensure alignment with sponsor expectations and academic deadlines.

- Mitigation strategies will focus on:
- Early API testing in sandbox environments
- Incremental LLM integration and prompt testing
- Strict adherence to read-only data policies
- Secure handling of API keys and sensitive data
- Scope control to prevent feature creep
- Clear escalation planning for unsupported chatbot responses

If a risk materializes, the team will document the issue, notify the project sponsor if necessary, and implement contingency actions to minimize project disruption.

9.2 Initial Risk List

Risk ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R1	Inaccurate or hallucinated LLM responses may provide incorrect or misleading information to guests.	Medium	High	Use Retrieval-Augmented Generation (RAG) with verified data, implement confidence thresholds, escalate low-confidence responses, and conduct structured testing.
R2	Integration issues with the StreamlineVRS API may delay development or limit functionality.	Medium	High	Begin integration early, use sandbox/testing environments, build mock API responses if needed, and enforce read-only access.
R3	External LLM provider (e.g., OpenAI) downtime, API limits, or cost constraints may disrupt service.	Medium	Medium	Monitor API usage, implement request limits, and design modular architecture to allow provider substitution if necessary.
R4	Data security or privacy breach involving reservation information or API keys.	Low	High	Use HTTPS for all communication, store no sensitive data persistently, secure API keys using environment variables, limit logging to non-sensitive data.
R5	Scope creep expands the project beyond defined POC boundaries (e.g., payments, booking creation).	Medium	Medium	Require sponsor approval for scope changes, maintain strict feature boundaries, document all change requests.
R6	Chatbot accuracy may not meet the 90% factual	Medium	High	Conduct controlled testing, refine prompts iteratively, improve retrieval logic, log unsupported inquiries for improvement.

	correctness objective.			
R7	Semester time constraints may limit testing, refinement, or feature completion.	High	High	Prioritize core features, follow sprint schedule, freeze new features after Milestone Two, allocate buffer time before final presentation.
R8	Team skill gaps in API integration or LLM implementation may slow development.	Medium	Medium	Use official documentation and tutorials, conduct early sandbox experiments, divide responsibilities based on strengths, seek instructor guidance when necessary.

Project Configuration and Data Management

10.1 Configuration Management

The reservation support chat platform will follow a simple but structured configuration management approach to ensure proper version control and ensure stability throughout development.

The project will use Git for version control and GitHub as the central code repository. The source code for the system and technical documentation can all be found in Github. During development, the team will follow a basic branching model to minimize merge conflicts. All changes will eventually be pushed to the main branch, assuming the branch is stable.

Project Process

11.1 Software Life Cycle Model

This project will follow the Agile Software Development Life Cycle (SDLC) focusing on short, iterative, and incremental development. This model allows for continuous refinement of the system. Development will occur in short 2-week sprints allowing the team to build core functionality, test RAG retrieval, improve prompts, and log data.

Glossary

Acronym	Meaning
AI	Artificial Intelligence
UI	User Interface
LLM	Large Language Model
POC	Proof of Concept
SRS	Software Requirements Specification
RAG	Retrieval-Augmented Generation
SDD	Software Design Document

Change Record

Date	Version	Changes
2/20/26	1.0	Initial Project Plan